



Dr Ravi Prakash

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AREAS OF EXPERTISE

Chemical Engineering

Expert in Multiphase Flow, CFD, Multiphysics Simulations, Microfluidics, Porous Media Transport, Wave-Structure Interaction, and Experimental Fluid Mechanics.

EDUCATION

PhD 2023

Indian Institute of Technology Roorkee

M.Tech. 2018

Indian Institute of Technology Roorkee

B.Tech 2015

Dr B R Ambedkar National Institute of Technology, Jalandhar

PROFESSIONAL EXPERIENCE

National Institute of Technology, Agartala 2025 - Present

Assistant Professor (Grade-II)

West Tripura

Loughborough University 2025 - 2026

Post-Doctoral Research Associate

Loughborough

University of Tokyo 2023 - 2024

Post-Doctoral Research Associate

Tokyo

RESEARCH PROJECTS

Smart Micro-emulsion Printing

Role: PI

Year: 2026 | Amount: 6960000

HONOURS AND AWARDS

International travel grant 2023

Awarded by: DST

MHRD Fellowship 2018

Awarded by: MHRD

PATENTS

Trapezoidal with fin as Float Design of Point Absorber Wave Energy Convertor

Inventor(s): Sumana Ghosh, Ravi Prakash, Ranjana Rathaur, Shivam Gupta, Deepak Kumar, Rajesh Agarwal, Sanjay Kumar Vatsa, Meghasham Khandige

Assignee: IIT Roorkee, ONGC | Published: 2023-05-18

A LOW COST MILI FLUIDIC DEVICE AND METHOD FOR PRODUCTION OF DOUBLE EMULSION

Inventor(s): Ravi Prakash,Pushpender, Prof. Sumana Ghosh

Assignee: IIT Roorkee | Published: 2026-05-11

A HEAVING POINT ABSORBER WAVE ENERGY CONVERTER FOR GENERATING ELECTRICITY FROM OCEAN ENERGY

Inventor(s): 1.SHIVAM GUPTA 2.RAVI PRAKASH 3.PROF. SUMANA GHOSH 4.PROF. MUKESH KUMAR PATHAK 5.BHARAT SITARAM MENDHE 6.DEEPAK KUMAR 7.PARMOD VAISHNAV 8.PRITI SARKAR

Assignee: IIT Roorkee, ONGC | Published: 2025-07-31

PUBLICATIONS

1. **Hydrodynamics and mass transfer in miniaturized helical coil for process intensification**

Chaudhary P.;Prakash R.;Kumar V.;Nigam K.D.P.;Ghosh S.

Chemical Engineering and Processing Process Intensification , Vol. 224 (2026) , pp.

2. **Integrating experiments numerical simulations and Bayesian optimisation to maximise drug permeability of hollow pyramidal microneedle arrays**

Ravi Prakash and Rahul Nadda and Abraham M. Abraham and Tahir Emre Yalcin and Ryan F. Donnelly and Diganta Bhusan Das

Chemical Engineering Journal , Vol. 538 (2026) , pp. 176974

3. **Incorporating fluid-structure interactions for modelling of pyramidal hollow microneedles for transdermal drug delivery**

4. **Numerical modelling for coupled flow, transport pharmacokinetic and pharmacodynamic of tablet-loaded hollow microneedle patch for transdermal drug delivery**
Ravi Prakash and Rahul Nadda and Diganta Bhusan Das and Ryan F. Donnelly
International Journal of Pharmaceutics , Vol. 686 (2025) , pp. 126309
5. **Experimental visualization of mass transfer from a slug bubble during co-current flow in a conventional channel**
Rohilla L.;Prakash R.;Verma R.K.
Chemical Engineering Science , Vol. 283 (2024) , pp.
6. **Experimental and theoretical study of thin film formation in a straight mini capillary**
Ravi Prakash and Sumana Ghosh
Chemical Engineering Journal , Vol. 452 (2023) , pp. 139578
7. **A numerical study on the effect of wall wettability on film formation in liquid-liquid slug flow**
Prakash, Ravi and Ghosh Sumana
Physics of Fluids , Vol. 35 (2023) , pp. 127117
8. **A numerical study on float design for wave energy converter**
Prakash, Ravi; Rathaur, Ranjana; Gupta, Shivam; Ghosh, Sumana; Kumar, Deepak; Agarwal, Rajesh; Vatsa, Sanjay Kumar; Khandige, Meghasham
Ocean Engineering (2022)
9. **Effect of jump in wettability on the hydrodynamics of liquid-liquid two-phase flow in a straight mini capillary tube**
Ravi Prakash and Sumana Ghosh
Chemical Engineering Journal , Vol. 435 (2022) , pp. 135194
10. **Effect of Bend Wettability on Hydrodynamics of Liquid-Liquid Two-phase Flow in Serpentine Mini Geometry**
Prakash, Ravi and Ghosh Sumana
Industrial \& Engineering Chemistry Research , Vol. 60 (2021) , pp. 3142-3155
11. **Liquid-liquid mass transfer in a serpentine miniature geometry- effect on pressure drop**
Prakash R.;Kumar Verma R.;Ghosh S.
Chemical Engineering Journal , Vol. 369 (2019) , pp. 489-497
12. **Biodiesel production in a serpentine minireactor—Effect of flow distribution**
Verma R.K.;Prakash R.;Mehta A.;Ghosh S.
International Journal of Energy Research , Vol. 43 (2019) , pp. 3461-3474